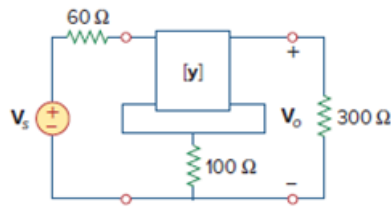


Problem

In the two-port of Fig. 19.113, let $y_{12} = y_{21} = 0$, $y_{11} = 2 \text{ mS}$, and $y_{22} = 10 \text{ mS}$. Find V_o/V_s .

Figure 19.113



Step-by-step solution

Step 1 of 3

Consider the following two port network.

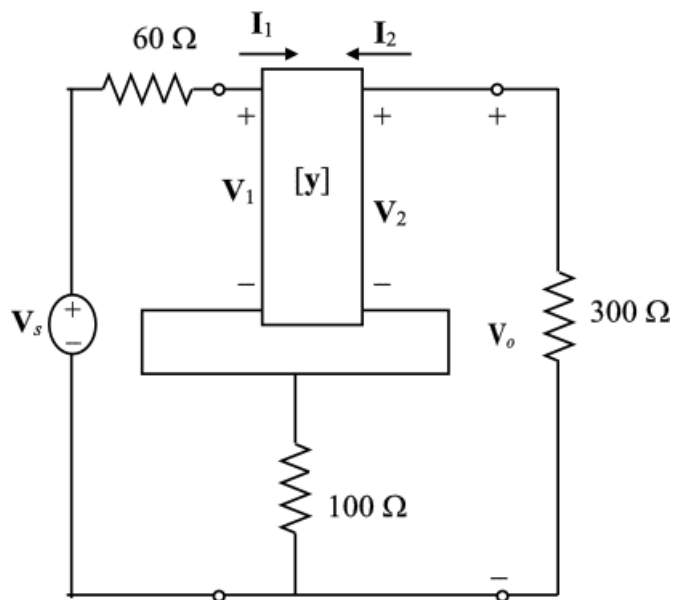


Figure 1

Step 2 of 3

The y parameters of the network are,

$$y_{11} = 2 \text{ mS}$$

$$y_{12} = y_{21} = 0$$

$$y_{22} = 10 \text{ mS}$$

Convert y parameters to z parameters using the following conversion formulae.

$$[\mathbf{z}] = \begin{bmatrix} \frac{y_{22}}{\Delta_y} & -\frac{y_{12}}{\Delta_y} \\ -\frac{y_{21}}{\Delta_y} & \frac{y_{11}}{\Delta_y} \end{bmatrix}$$

Determine Δ_y .

$$\begin{aligned} \Delta_y &= y_{11}y_{22} - y_{12}y_{21} \\ &= (2 \times 10^{-3})(10 \times 10^{-3}) - 0 \\ &= 2 \times 10^{-5} \end{aligned}$$

Determine $[\mathbf{z}]_a$

$$\begin{aligned} [\mathbf{z}] &= \begin{bmatrix} \frac{10 \times 10^{-3}}{2 \times 10^{-5}} & -\frac{0}{2 \times 10^{-5}} \\ -\frac{0}{2 \times 10^{-5}} & \frac{2 \times 10^{-3}}{2 \times 10^{-5}} \end{bmatrix} \\ &= \begin{bmatrix} 500 & 0 \\ 0 & 100 \end{bmatrix} \end{aligned}$$

Determine the z parameters of the series combination of y-network and 100 ohms resistance.

$$\begin{aligned} [\mathbf{z}] &= [\mathbf{z}]_a + \begin{bmatrix} 100 & 100 \\ 100 & 100 \end{bmatrix} \\ &= \begin{bmatrix} 500 & 0 \\ 0 & 100 \end{bmatrix} + \begin{bmatrix} 100 & 100 \\ 100 & 100 \end{bmatrix} \\ &= \begin{bmatrix} 600 & 100 \\ 100 & 200 \end{bmatrix} \end{aligned}$$

Expand the z parameters.

$$\mathbf{V}_1 = 600\mathbf{I}_1 + 100\mathbf{I}_2$$

$$\mathbf{V}_2 = 100\mathbf{I}_1 + 200\mathbf{I}_2$$

$$\mathbf{I}_1 = \frac{1}{100}(\mathbf{V}_2 - 200\mathbf{I}_2)$$

Apply Kirchhoff's voltage law at the input port.

$$\mathbf{V}_s = \mathbf{V}_1 + 60\mathbf{I}_1$$

Substitute $600\mathbf{I}_1 + 100\mathbf{I}_2$ for $\mathbf{V}_1$.

$$\begin{aligned}\mathbf{V}_s &= 600\mathbf{I}_1 + 100\mathbf{I}_2 + 60\mathbf{I}_1 \\ &= 660\mathbf{I}_1 + 100\mathbf{I}_2\end{aligned}$$

Substitute $\frac{1}{100}(\mathbf{V}_2 - 200\mathbf{I}_2)$ for $\mathbf{I}_1$.

$$\begin{aligned}\mathbf{V}_s &= 660\left[\frac{1}{100}(\mathbf{V}_2 - 200\mathbf{I}_2)\right] + 100\mathbf{I}_2 \\ &= 6.6\mathbf{V}_2 - 1220\mathbf{I}_2 \\ &= 6.6\mathbf{V}_o - 1220\mathbf{I}_2\end{aligned}$$

$$\mathbf{I}_2 = \frac{6.6\mathbf{V}_o - \mathbf{V}_s}{1220}$$

Apply Kirchhoff's voltage law at the output port.

$$\mathbf{V}_2 = \mathbf{V}_o = -300\mathbf{I}_2$$

Substitute $\frac{6.6\mathbf{V}_o - \mathbf{V}_s}{1220}$ for $\mathbf{I}_2$.

$$\mathbf{V}_o = (-300)\left(\frac{6.6\mathbf{V}_o - \mathbf{V}_s}{1220}\right)$$

$$1220\mathbf{V}_o = -1980\mathbf{V}_o + 300\mathbf{V}_s$$

$$3200\mathbf{V}_o = 300\mathbf{V}_s$$

$$\frac{\mathbf{V}_o}{\mathbf{V}_s} = \frac{300}{3200}$$

$$\frac{\mathbf{V}_o}{\mathbf{V}_s} = 0.09375$$

Thus, the ratio $\frac{\mathbf{V}_o}{\mathbf{V}_s}$ is 0.09375.